

Weakness, Toroidal Topology, and OOD Generalization in Path-Integration RNNs: A Modal Gate Sweep

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Research-Derived Experiments · Paper A

Abstract

This paper tests a strong registered claim: that **weakness**, a scalar measuring how many translations a learned code remains compatible with, governs whether a path-integration RNN forms a toroidal population manifold and whether that topology mediates larger-arena generalization. A 320-network Modal H100 sweep gives a mixed, informative answer. Full translation augmentation reliably induces toroidal codes (G1 pass; torus match 0.734), spectral concentration tracks weakness (G5 pass; $\rho=+0.635$), and the causal condition contrast is large (G6 pass; OOD 0.949 vs 0.484 for no augmentation). But the central triangle claim fails: weakness only weakly predicts toroidal score (G2 fail; $\rho=+0.197$), weakness does not beat the best classical OOD baseline by the preregistered $2\times$ margin (G3 fail; weakness $\rho=+0.617$, loss $\rho=+0.652$), and topology does not mediate the weakness→OOD relation (G4 fail). The honest conclusion is narrower and stronger for being bounded: translation augmentation causally produces a toroidal, OOD-generalizing spatial code, but this sweep does not establish weakness as the governing scalar of toroidal topology.

1. Question

Grid cells in mammalian entorhinal cortex form periodic spatial codes whose population activity lies on a torus. RNNs trained for path integration can rediscover similar codes. The preregistered question here is whether the program's substrate-general scalar, **weakness**, predicts that topological event. Weakness is the mean held-out R^2 with which one linear operator can transform a population code under wrapped translations: a high-weakness code behaves as though it has learned the translation group.

The strong version predicted a triangle: high weakness, low Fourier participation ratio (few aligned irreducible representations), clean toroidal topology, and larger-arena OOD generalization should be measurements of one event. The sweep below tests that triangle without moving the goalposts.

2. Method

We trained 320 velocity-driven path-integration RNNs on Modal H100 workers: five training conditions, two recurrent architectures, and 32 seeds. Each network predicts a place-cell code from velocity sequences. Conditions vary the augmentation regime: full translations, partial translations, random shifts, no augmentation, and a wrong-group control. For each trained code we measure weakness, persistent-homology toroidal score, Fourier participation ratio, and decoding accuracy in held-out arenas of scale 1.0, 1.25, 1.5, and 2.0. The largest arena is the preregistered OOD metric.

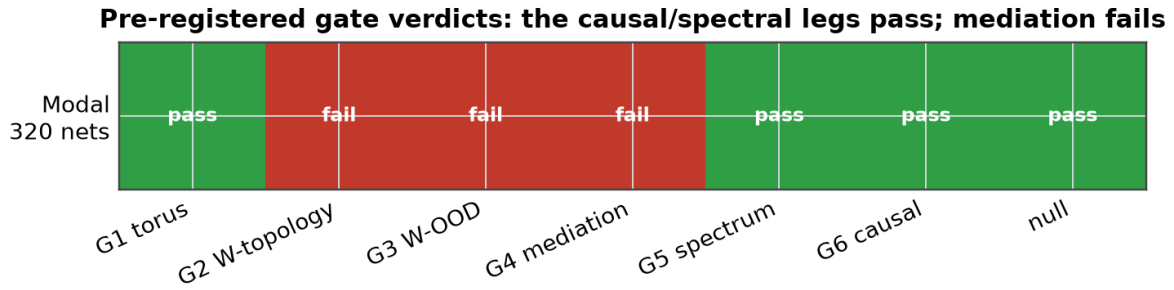


Figure 1. Gate verdicts from the preregistered Modal sweep. The causal and spectral legs pass; the topology-governing and mediation claims fail.

3. Results

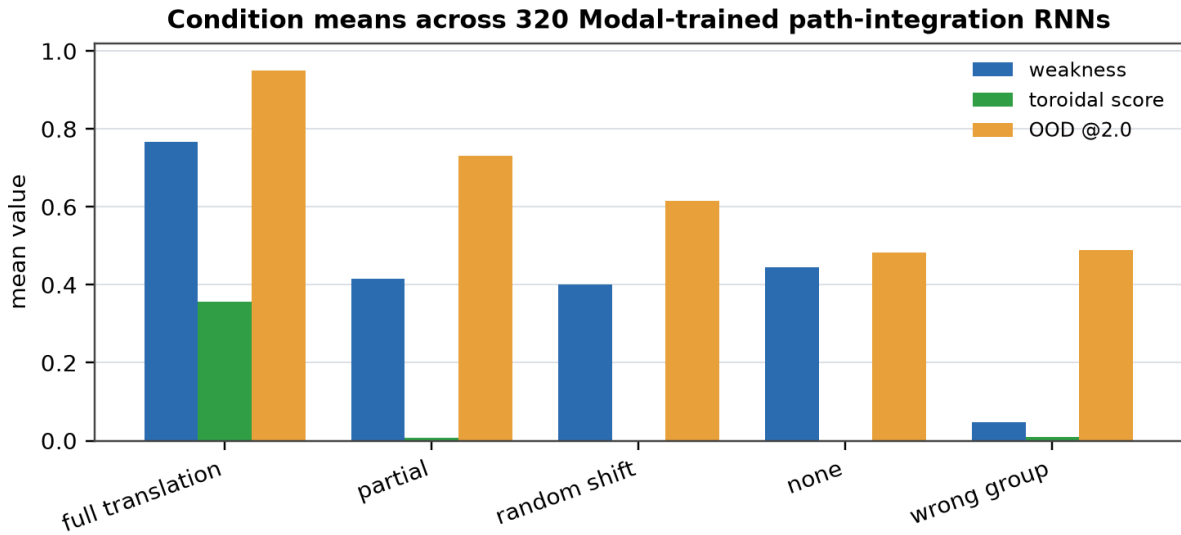


Figure 2. Condition means. Full translation is the only condition that reliably forms a torus and preserves OOD decoding at arena scale 2.0.

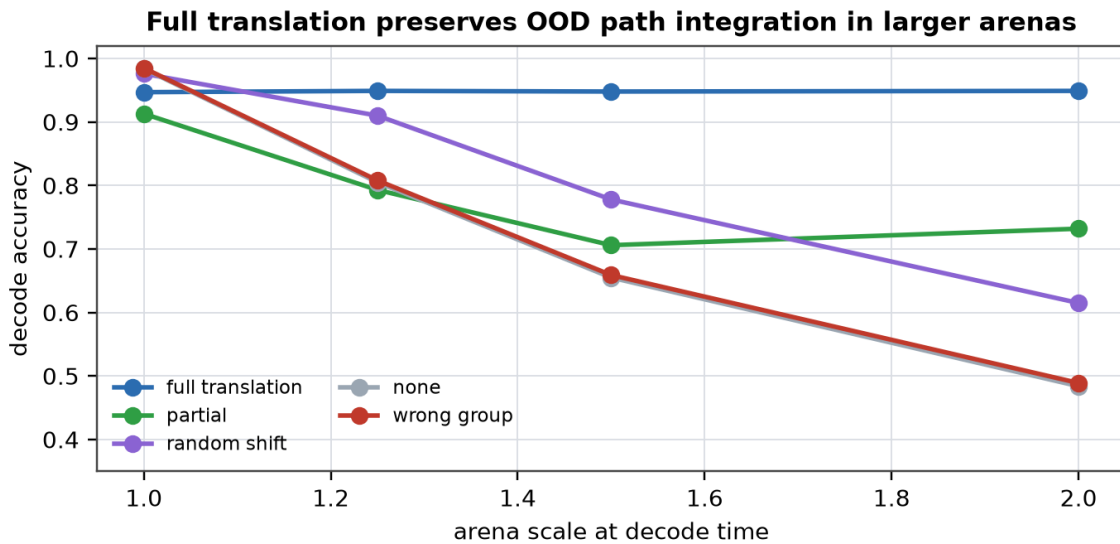


Figure 3. Larger-arena OOD curves. Full translation remains stable as the arena doubles; controls degrade sharply.

Gate	Result	Verdict
G1 manifold recovered	full-translation torus match = 0.734	pass
G2 weakness $\leftrightarrow$ topology	$\rho = +0.197$; loss $\leftrightarrow$ topology $\rho = +0.431$	fail
G3 weakness $\leftrightarrow$ OOD	$\rho = +0.617$; loss $\leftrightarrow$ OOD $\rho = +0.652$	fail
G4 topology mediates	partial $\rho = +0.623$; drop = -0.009	fail
G5 spectral leg	$\rho(\text{weakness}, -\text{Fourier PR}) = +0.635$	pass
G6 causal contrast	topology 0.357 vs ~ 0 ; OOD 0.949 vs 0.484	pass
wrong-group null	$\rho = 0.000$ with tie-aware Spearman	pass

Table 1. Pre-registered gate verdicts. The sweep confirms torus formation under the causal intervention but rejects the strongest mediation story.

4. Interpretation

The result should not be polished into a full confirmation. It is better than that: it tells us exactly which parts of the theory survive scale. Translation augmentation is a real causal intervention for toroidal topology and larger-arena generalization. Weakness is still a useful spectral and OOD-associated quantity. But toroidal topology is not the mediator of the weakness $\rightarrow$ OOD relationship in this harness, and weakness is not yet the single scalar that controls torus formation.

A plausible reading is that the augmentation condition itself is the strongest causal variable: it directly supplies the group structure needed for both topology and OOD. Weakness measures part of that event, especially its spectral footprint, but it is not sufficiently specific to explain all topology variance after training loss and condition effects are present. Future work should test mediation within a narrower family of already well-trained toroidal codes and use tie-aware rank statistics from the outset.

5. Limitations

This remains an artificial RNN path-integration task, not a biological replication. The topology metric uses binned hidden activity and Vietoris-Rips persistent homology; both can be sensitive to sampling density. The wrong-group null in the first Modal summary was miscomputed by a rank helper that did not average ties; the corrected tie-aware Spearman is reported here and in the result report. The raw cell measurements did not change.

References

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